WEEK

6

Stochastic Programming : Demand is Uncertain

Why the Deterministic Model is Not Enough

Imagine you are managing inventory for a chai stall outside a college campus. On most days you sell around 80 cups. But on exam days you sell 140. On rainy days only 30. You have to decide how many cups to prepare before the day begins, before you know which kind of day it will be.

The honest answer is that demand is uncertain. You do not know the exact number. You know something about what it might be. Maybe you know from experience that demand is usually between 30 and 140, with most days clustered around 80. That knowledge can be captured mathematically using scenarios.

What Changes in the Model

Deterministic	Stochastic
Fixed known numbers	Random
o_w[t], o_r[t], I_w[t], I_r[t]	Same variables but now indexed by scenario too
Minimise total cost	Minimise expected cost across all scenarios
Balance equations for each period	Balance equations for each period AND each scenario
One optimal plan	One plan that works well on average across all futures

What is a Scenario?

A scenario is one possible version of how demand plays out across all time periods. It is a complete sequence of demand values from period 0 to period T minus 1, representing one particular future that might happen.

You do not have just one scenario. You have a collection of them, each representing a different possible future. Each scenario has a probability - how likely that future is. All probabilities must add up to 1.

Three Manual Scenarios to Start

Before generating scenarios randomly, look at three written out explicitly. This is a simplified version of the model with T = 5 periods and one retailer only, so you can see the structure clearly.

Scenario	Probability	t=0	t=1	t=2	t=3	t=4
Scenario 1 (Low demand)	1/3 (33%)	18	22	20	25	19
Scenario 2 (Normal)	1/3 (33%)	28	32	30	35	29
Scenario 3 (High demand)	1/3 (33%)	40	45	42	48	38

First-Stage and Second-Stage Decisions

1

First-stage decisions
These are the order quantities o_w[t]. You place these orders before the period begins, before you see actual demand. They must be the same regardless of which scenario happens. The solver chooses one set of order quantities that works well across all scenarios.

2

Second-stage decisions
These are the inventory levels I_w[t][s] and I_r[t][s]. After demand is revealed in each scenario, the inventory levels are determined by the balance equations. They can be different for each scenario because different demands lead to different inventory outcomes.

The New Objective

If scenario s has probability p[s] and the total cost under scenario s is Cost[s], then expected cost is:

```
Expected Cost  =  Σ_s  p[s]  ×  Cost[s]
```

Sum over all scenarios s, each weighted by its probability

In the LP this becomes:

```
Minimise  Σ_t [ c × o_w[t] ]  +  Σ_s Σ_t  p[s] × [ h_w ×
      I_w[t][s]  +  h_r × I_r[t][s] ]
```

Ordering cost has no scenario index. Holding costs are averaged across scenarios.

Part A = Three Manual Scenarios

Start here. code using the three scenarios written out .The model is simplified to one retailer and T = 5 periods.

```
# Stochastic LP with 3 manual scenarios
# Simplified: 1 warehouse, 1 retailer, T = 5, S = 3

import pulp

T  = 5
S  = 3
LT = 1

scenarios = [
    [18, 22, 20, 25, 19],   # Scenario 0: low demand
    [28, 32, 30, 35, 29],   # Scenario 1: normal demand
    [40, 45, 42, 48, 38],   # Scenario 2: high demand
]
prob = [1/3, 1/3, 1/3]     # each scenario equally likely

# Costs and starting inventory
h_w     = 0.5
h_r     = 1.0
c_order = 2.0
p_short = 8.0    # shortage penalty: cost per unit of unmet demand
I_w0    = 80
I_r0    = 20

# Solve

# the first-stage order quantities

# second-stage outcomes per scenario
```

Part B = Ten Generated Scenarios

Now extend to S = 10 scenarios generated randomly using numpy. The structure is identical to Part A. The only difference is how the scenarios are created and the model uses two retailers .

Complete both codes.

Deliverable

Complete both codes

Compare with deterministic

Increase shortage penalty

Increase demand variability